Notations :

1.Options shown in green color and with ✔ icon are correct.

2.Options shown in red color and with ✖ icon are incorrect.

Change Font Color :	No
Change Background Color :	No
Change Theme :	No
Help Button :	No
Show Reports :	No
Show Progress Bar :	No

Speech Technology

Section Id :	640653134553
Section Number :	1
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	3
Number of Questions to be attempted :	3
Section Marks :	25
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653374463
Question Shuffling Allowed :	No

Question Number : 1 Question Id : 6406532040137 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : SPEECH TECHNOLOGY (PEN AND

PAPER EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406536642607. ✔ Yes

6406536642608. ✖ No

Question Number : 2 Question Id : 6406532040138 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

Instructions:

All **option-based questions are MCQs,** and all **fill-in-the-blank questions are short answer questions.**

For Questions **3 (1 to 14),** answer all questions.

• In **MCQs,** you must write the **correct option** and also the **complete text of that option.**
• You must then provide a **detailed explanation** of why that specific option is the correct answer.
• In **short answer questions,** write the answer clearly with proper explanation wherever needed.
• In all **problem-solving or example-based questions,** present the solution with **all intermediate steps clearly shown.**
• Step-wise explanation is **compulsory** for all such questions.
• The **final answer** must be written clearly at the end.

Options :

6406536642609. ✔ Instructions has been mentioned above.

6406536642610. ✖ This Instructions is just for a reference & not for an evaluation.

Sub-Section Number :	2
Sub-Section Id :	640653374464
Question Shuffling Allowed :	No

Question Number : 3 Question Id : 6406532040139 Question Type : MCQ

Correct Marks : 25

Question Label : Multiple Choice Question

1. (1 point) A key advantage of continuous HMM over discrete HMM in speech recognition is:
   (a) No need for feature extraction
   (b) Direct modeling of real-valued acoustic features
   (c) No training data required
   (d) No state transitions required

2. (1 point) In a typical 3-state phoneme HMM, self-loops are mainly used to model:
   (a) Vocabulary growth
   (b) Speaker identity
   (c) Variable duration
   (d) Word frequency

3. (1 point) If attention weights shift backward in time during decoding, what does it violate?
   (a) CTC assumption
   (b) Vocabulary constraint
   (c) Encoder out
   (d) Monotonic alignment

4. (1 point) Fill in the blank: The predictor in RNN-T is similar to a language model, since it conditions on _____.

5. (1 point) The fine-tuning step of wav2vec 2.0/HuBERT using a simple linear layer for ASR typically uses:
   (a) Connectionist Temporal Classification (CTC) loss
   (b) Cross-entropy loss with phoneme labels
   (c) Attention loss
   (d) Contrastive loss

6. (1 point) Which statement is **false** about CTC?
   (a) CTC assumes conditional independence between output labels at different time steps.
   (b) CTC decoding often uses greedy or beam search.
   (c) CTC explicitly learns alignments between input and output during training.
   (d) CTC introduces a special blank token.

7. (1 point) A CNN has 3 layers with strides:
   Layer 1: stride = 4
   Layer 2: stride = 4
   Layer 3: stride = 5
   What is the total downsampling factor?
   (a) 40
   (b) 60
   (c) 80
   (d) 100

8. (1 point) At a specific node (t, u) in the RNN-T lattice, the Joiner network receives h_t^{enc} and h_u^{pred}. If the Joiner outputs a probability for the "blank" symbol, what does a transition to the next state represent?
   (a) A move to $(t, u+1)$
   (b) A move to $(t+1, u+1)$
   (c) Termination of the sequence.
   (d) A move to $(t+1, u)$

9. (1 point) In an attention-based ASR system, suppose the attention weights are uniform across all encoder time steps. What effect does this have on the alignment between input speech frames and output tokens?

   (a) Good alignment

   (b) Poor alignment

   (c) Faster decoding

   (d) No decoding

10. (1 point) Suppose GPT predicts probabilities for the next token:
    "cat" $\rightarrow$ 0.7,
    "dog" $\rightarrow$ 0.2,
    "car" $\rightarrow$ 0.1
    If the true token is "cat", what is the cross-entropy loss?

    (a) -log(0.7)

    (b) -log(0.2)

    (c) -log(0.1)

    (d) -log(0.3)

11. (2 points) In automatic speech recognition (ASR), how does the Word Error Rate (WER) typically vary across CTC, RNN-T, and AED (Attention-based Encoder-Decoder) models, assuming similar training data and model capacity?

    (a) WER(CTC) < WER(RNN-T) < WER(AED)

    (b) WER(RNN-T) < WER(AED) < WER(CTC)

    (c) WER(CTC) = WER(RNN-T) = WER(AED)

    (d) WER(AED) < WER(RNN-T) < WER(CTC)

12. (3 points) Find the probability of the sentence

$$< s > \textit{ She likes bananas } < s >$$

with a **bigram** language model trained on the following corpus:

- $< s >$ She likes apples $< s >$
- $< s >$ He likes oranges $< s >$
- $< s >$ She eats apples $< s >$
- $< s >$ Ram likes bananas $< s >$

Note: $< s >$ is a special word to include start and end words of a sentence. For example, ($< s >$, That) is a valid bi-gram. Bi-gram means two consecutive words considered at a time.

13. (5 points) Consider an RNN-T model with input length $T = 2$ (two acoustic frames) and target sequence $Y = (A, B)$ of length $U = 2$. An alignment corresponds to a path in the (t, u) lattice starting from $(0, 0)$ and ending at $(T, U) = (2, 2)$, where:

Task: List all possible valid alignments that produce the output sequence "A B". For each alignment, write down the sequence of actions (blanks and symbols) along the path.

14. (5 points) Suppose the target sequence is AB. The input has 3 time steps, and the alphabet is $\{A, B, _\}$ (where _ = blank). The model outputs the following frame-level probabilities:

Time step	$P(A)$	$P(B)$	$P(_)$
$t1$	0.6	0.2	0.2
$t2$	0.1	0.7	0.2
$t3$	0.1	0.6	0.3

List all valid CTC alignment paths of length 3 that collapse to AB, compute their probabilities, and then compute the total probability of the sequence AB under CTC.
Note: Alignments can include blank at start and end

(a) 0.33

(b) 0.45

(c) 0.46

(d) 0.50

Options :

6406536642611. ✔ I have answered this question in the offline paper

6406536642612. ✖ I have skipped this question

6406536642613. ✖ This question does not apply to me